

PROJECT LAB REPORT

# Network Reconnaissance with Nmap

Scanning, service discovery, and security risk analysis on an isolated local virtual machine

```
$ nmap -sV -O 192.168.56.101
```

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**Tools:** Nmap · Linux Terminal · VirtualBox VM (Kali / Ubuntu)

*⚠ Authorized lab environment only — self-owned VM, no external targets*



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# What this lab sets out to do

- Stand up an isolated local VM as a safe, legal scan target
- Install and run Nmap to perform host discovery and port scanning
- Layer in service-version detection (-sV) and OS fingerprinting (-O)
- Catalog every open port, its service, and its security implications
- Produce structured, reviewable documentation of the findings

## DELIVERABLES

- README.md — Nmap overview, ethics, setup
- nmap\_scan\_results.txt — structured findings
- Terminal screenshots for each scan
- Per-port risk write-up

# What is Nmap?

Nmap (“Network Mapper”) is a free, open-source utility for network discovery and security auditing. It sends crafted packets to target hosts and interprets the responses to determine which hosts are reachable, which ports are open, what services and versions are running, and — with enough signal — what operating system a host is likely running.

1

## Host Discovery

Determine which IP addresses on a network are live and responding

2

## Port Scanning

Identify open, closed, and filtered TCP/UDP ports on a target

3

## Service / Version Detection

Fingerprint the software (and version) listening on each open port

4

## OS Detection

Infer the target's operating system from TCP/IP stack behavior

# Why network scanning matters

- Attackers scan first — reconnaissance is step one of nearly every intrusion, so defenders need to see what an attacker would see
- You can't secure what you can't see: unknown open ports and forgotten services are a leading cause of breaches
- Regular scanning turns an unknown attack surface into an inventoried, monitored one

## DEFENSIVE VALUE

### Asset visibility

Know exactly which services are exposed

### Misconfiguration checks

Catch services left on by mistake

### Patch prioritization

Version data flags outdated, vulnerable software

### Compliance

Many standards require periodic scanning

# Tools & lab environment

## Nmap

Open-source scan engine — host discovery, port, version, and OS scans

## Hypervisor

VirtualBox (or VMware) hosting an isolated virtual network

## Attacker VM

Kali Linux — Nmap pre-installed, purpose-built for security testing

## Target VM

A second local VM (e.g. Ubuntu Server) you fully own and control

## SUGGESTED NETWORK LAYOUT

Both VMs run on a Host-Only or Internal network adapter (not Bridged) so the scan traffic never leaves your machine. Note the target VM's IP with `ip a / ifconfig` before scanning — that address is what you pass to Nmap.

# Installing Nmap

## Kali Linux

Nmap ships preinstalled — verify with:

```
$ nmap --version
```

## Debian / Ubuntu

Install via APT:

```
$ sudo apt update && sudo apt install nmap -y
```

## Fedora / RHEL

Install via DNF:

```
$ sudo dnf install nmap -y
```

## Windows (PowerShell)

Download the installer from [nmap.org](https://nmap.org), or use winget:

```
$ winget install Insecure.Nmap
```

*Document the exact commands and output you used in README.md — screenshots included.*

# Ethical use guidelines



**Only scan systems you own or have explicit written permission to test.** Unauthorized scanning of third-party or production systems can violate laws such as the U.S. Computer Fraud and Abuse Act and equivalent statutes elsewhere.

- Always target a local VM you control — never an external or production system
- Keep scan traffic on an isolated Host-Only / Internal virtual network
- Get written authorization before scanning any system that isn't purely yours
- Use findings defensively — to fix exposures, not to exploit them
- Handle scan results as sensitive data; don't publish real target details publicly

# Three scans, three layers of insight

01

## Basic Scan

Default SYN scan of the top 1,000 TCP ports  
— tells you what's open

```
$ nmap [target]
```



02

## Service Version Scan

Probes each open port to identify the  
running software and version

```
$ nmap -sV [target]
```



03

## OS Detection Scan

Analyzes TCP/IP stack behavior to fingerprint  
the operating system

```
$ nmap -O [target]
```



## Basic scan — nmap [target IP]

*Illustrative template — replace with your VM's actual output*

```
Starting Nmap 7.94 ( https://nmap.org )  
Nmap scan report for 192.168.56.101  
Host is up (0.00042s latency).  
Not shown: 996 closed tcp ports (reset)
```

| PORT    | STATE | SERVICE      |
|---------|-------|--------------|
| 22/tcp  | open  | ssh          |
| 80/tcp  | open  | http         |
| 139/tcp | open  | netbios-ssn  |
| 445/tcp | open  | microsoft-ds |

```
Nmap done: 1 IP address (1 host up) scanned in 2.31 seconds
```

*Capture the real terminal output as a screenshot for the GitHub repo and paste the raw text into nmap\_scan\_results.txt.*

# Service version scan — nmap -sV

*Illustrative template — replace with your VM's actual output*

| PORT    | STATE | SERVICE      | VERSION              |
|---------|-------|--------------|----------------------|
| 22/tcp  | open  | ssh          | OpenSSH 8.9p1 Ubuntu |
| 80/tcp  | open  | http         | Apache httpd 2.4.52  |
| 139/tcp | open  | netbios-ssn  | Samba smbd 4.15.13   |
| 445/tcp | open  | microsoft-ds | Samba smbd 4.15.13   |

## OS detection scan — nmap -O

*Illustrative template — replace with your VM's actual output*

```
Running: Linux 5.X
OS CPE: cpe:/o:linux:linux_kernel:5
OS details: Linux 5.4 - 5.15
Network Distance: 1 hop
```

```
Aggressive OS guesses:
Linux 5.4 (92%)
Linux 5.10 (89%)
```

### READING OS DETECTION

- OS detection relies on subtle TCP/IP stack fingerprints — treat results as a confident guess, not certainty
- Confidence percentages matter: 92% is reliable, 40% is not
- Firewalls and NAT can distort or block detection entirely
- Requires root/administrator privileges to run

# Open ports at a glance

| PORT    | SERVICE     | WHAT IT DOES                        | RISK   |
|---------|-------------|-------------------------------------|--------|
| 22/tcp  | SSH         | Encrypted remote shell access       | MEDIUM |
| 80/tcp  | HTTP        | Unencrypted web server traffic      | MEDIUM |
| 139/tcp | NetBIOS-SSN | Legacy Windows file/printer sharing | HIGH   |
| 445/tcp | SMB         | Windows/Samba file sharing protocol | HIGH   |

# Per-port risk analysis

## 22 / SSH

**MEDIUM**

Encrypted, but a prime brute-force target. Weak passwords, default credentials, or outdated OpenSSH versions turn this into an easy foothold.

## 80 / HTTP

**MEDIUM**

Traffic is unencrypted — credentials and session data can be sniffed. Outdated web software adds exposure to known CVEs.

## 139 / NetBIOS-SSN

**HIGH**

Legacy Windows networking protocol with a long history of exploits (e.g. worms). Rarely needs to be exposed beyond a trusted LAN.

## 445 / SMB

**HIGH**

File-sharing protocol tied to critical past exploits (EternalBlue/WannaCry). Unpatched or misconfigured SMB is a top ransomware entry point.

# Documentation & repo structure

## REPO LAYOUT

```
nmap-scan-lab/  
├─ README.md  
├─ nmap_scan_results.txt  
└─ screenshots/  
    ├─ basic_scan.png  
    ├─ service_version_scan.png  
    └─ os_detection_scan.png
```

## README.md MUST COVER

- What Nmap is and what it's used for
- Why network scanning matters for security
- Ethical use guidelines and legal boundaries
- Installation steps for your OS
- How to reproduce each of the three scans

# Key takeaways



## Visibility first

Scanning turned an assumed-secure VM into a documented, inventoried attack surface



## Version data drives action

Knowing exact service versions is what makes a finding actionable, not just informational



## Legacy services are the risk

The highest-risk findings here were older, unnecessary protocols (NetBIOS/SMB) — prime candidates to disable



## Ethics is not optional

Every scan in this project stayed inside an isolated, self-owned lab environment

*Next: schedule recurring scans and re-check this VM after disabling unused services.*